

Karan Bajpai

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EDUCATION

B.S. in Computer Science - Pacific University, Forest Grove, OR August 2020 - May 2024
GPA: 3.7 **Awards:** Dean's List, Recipient of Future Connect Scholarship **Honors:** Magna Cum Laude
General Education - Portland Community College, Portland, OR September 2018 - June 2020
GPA: 3.90 **Awards:** President's List, Dean's List, Recipient of Future Connect Scholarship
Relevant Courses: Data Structures / Object Oriented Programming / Operating Systems / Computer Networking / Computer Security / Web Development / Mobile Development / Object Oriented Design / Algorithms / Databases

SKILLS

Robotics Hardware:

- Basic hardware tools including oscilloscopes and VOMs / Hardware protocols including SPI, I2C, and PWM / Sensors including IMU, LIDAR, GPS, and Depth Cameras / Actuators including servos and PWM motor control / UI elements including gamepads, audio output, text displays, and LEDs / Single Board computers including Raspberry Pi & Arduino

Robotics Software - ROS2:

- Low level integration to ROS topics / Located and coded nodes for camera, lidar, 9-axis IMU, DC motor control, gamepad / Launch files / Gazebo simulation

Embedded Linux:

- Process creation and management / Simple interprocess communication / File I/O

Programming:

- Languages (C, C++, Python, SQL, Java, JavaScript) / Advanced Object Oriented Design / Event driven programming / TCP client & server programming / Tools including Git, Visual Studio Code, & Wireshark

PROJECTS

Online Bank System - C++, File I/O, UML

- Designed a backend system for managing account logic, transactions, and file persistence
- Used OOP principles to model checking / savings operations with error handling
- Strengthened structured system thinking, data integrity, and embedded-safe file management

Chess-playing Robot - C++, Python, Arduino Mega, Raspberry Pi

- Engineered a robotic arm that plays chess by communicating with Stockfish AI API
- Programmed precise motor control using Arduino and system logic in Python on Raspberry Pi
- Showcased multi-device embedded coordination and real-time physical actuation

Shell Emulator - C, Linux System Programming

- Built a UNIX-style shell with support for piping, redirection, and background processes
- Used system calls to handle process creation and inter-process communication
- Reinforced embedded Linux and OS-level control critical for low-resource environments

TCP Socket Server - C, Linux, Wireshark

- Created a TCP client-server system capable of persistent data packet exchange
- Implemented custom protocol logic and validated integrity using Wireshark
- Applied low-level networking knowledge relevant to embedded device communication

Work Experience

Analog Devices, Inc - Mechatronics Technician II, Equipment Engineering

August 2025 - Present

- Programmed and tuned collaborative robots using TMFlow / TMScript, adjusting joint angles to optimize pick-and-place operations
- Configured and calibrated robot vision systems for landmark detection, enabling precise localization and automated handling with end effectors
- Modified Python scripts to interface with robots and automate control tasks
- Configured, troubleshooted, and maintained industrial wireless networks, including Phoenix Contact WLAN 1120 radios, IP/Wi-Fi settings, and lab Wi-Fi extenders for robot connectivity.
- Performed electronics assembly, including soldering, wiring, and crimping, following schematics and diagrams.
- Operated autonomous robots in a semiconductor cleanroom environment, using MobilePlanner software to move robots across the fab safely.
- Coordinated with global robotics support teams to diagnose and resolve automation system issues, ensuring uninterrupted production.